Discrete Mathematics (CSC 1204)

Chapter 1

1.1 Propositional Logic

Key Terms

- **Logic:** Logic is the discipline that deals with the methods of reasoning.
  - Logic is the basis of all mathematical reasoning
  - The rules of logic specify the meaning of mathematical statements

- **Propositional Logic**: The area of logic that deals with propositions is called the propositional logic.

Key Terms

- **Proposition:** A ***proposition*** is a declarative statement that's either **TRUE** or **FALSE**, but not both.

- Statements that are not propositions *include*
  - questions
  - commands

Key Terms

- **Propositional variable:** A variable that represents a proposition. The conventional letters used for propositional variables are *p, q, r, s, t,..*
- **Compound proposition:** A proposition constructed by combining two or more propositions using *logical operators* (*logical connectives*)
- **Logical Operators**: Operators used to combine propositions
- **Truth Value**: **The truth value** of a proposition is **true**, denoted by **T**, if it is a true statement and **false**, denoted by **F**, if it is a false statement. **Truth Value ==>** Either **True** or **False**
- **Truth Table:** A table displaying the truth values of propositions.

Proposition: Examples

Proposition		Not Proposition
3 + 2 = 32	false	Bring me coffee!
3 + 2 = 5	true	3 + 2
CSC 1204 is Katrina's favorite class		CSC 1204 is her favorite class
Every cow has four legs		Do you like Cake?

Proposition: More Examples

- Which of the following are propositions? What are the truth values of those that are propositions?

  a. Read this carefully.

  b. 2 + 2 = 4.

  c. Comilla is the capital of Bangladesh.

  d. What time is it?

Solution.

a. Not a proposition (because it's a command).

b. A proposition with truth value (T).

c. A proposition with truth value (F).

d. Not a proposition (because it's a question).

Logical Operators

- Logical operators ==> **unary**, **binary**
- **Unary**:
  - Negation
- **Binary**
  - Conjunction
  - Disjunction
  - Exclusive OR
  - Conditional/Implication
  - Biconditional

Logical Operators: Symbols & Usage

Operator	Symbol	Usage
Negation	$\neg$	NOT
Conjunction	$\wedge$	AND
Disjunction	$\vee$	OR
Exclusive or	$\oplus$	XOR
Conditional	$\rightarrow$	if, then
Biconditional	$\leftrightarrow$	iff

Propositional Logic - **Negation**

- Let *p* be a proposition. The *negation* of *p*, denoted by $\neg p$ (or $\bar{p}$), is the statement "It is not the case that *p*."
- The proposition $\neg p$ *is read "not p"*
- The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Propositional Logic - **Negation**

- Negation just turns *a **false*** proposition to ***true*** and the opposite for a true proposition.

- Example1: P: I am going to town

  ¬ P: I am not going to town; or,

  It is not the case that I am going to town

- Example2: *p* : "23 = 15 +7"

  p happens to be false, so ¬ *p* is true.

Truth table for Negation of a Proposition

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TABLE 1 The Truth Table for the Negation of a Proposition.

p	$\neg p$
T	F
F	T

Negation: Another Example

- Example: Find the negation of the proposition
  "Today is Friday."

- **Solution**: The negation is
  "It is not the case that today is Friday", or
  "Today is not Friday" , or
  "It is not Friday today."

Conjunction

- Let p and q be propositions. The *conjunction* of p and q, denoted by $p \wedge q$, is the proposition "p and q."
- The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.
- Conjunction corresponds to English "**AND**".

- Example: Liana is curious AND clever.

Truth Table for Conjunction

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TABLE 2 The Truth Table for the Conjunction of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Number of Combination
2 to the power n
2 to the power 2 (p,q)
= 4

Conjunction: Example

- **Example:** *p*: ‘I am going to town’

  q: ‘It is going to rain’

  $p \wedge q$: ‘I am going to town and it is going to rain.’

  Note: **Both *p* and *q*** must be true to $p \wedge q$ be **true**

Conjunction: Another Example

p = "Bill was the president."

q = "Hilary was the president."

r = "Obama was the president."

- Assuming *p* and *r* are true, while *q* false.

Out of $p \wedge q$, $p \wedge r$, $q \wedge r$, only $p \wedge r$ is **true**.

Question: what is the truth value of $p \wedge q \wedge r$?

Answer: **False** (because truth value of *q* is false)

Conjunction

- **Conjunction** is a ***binary*** operator in that it operates on two propositions when creating compound proposition.
- On the other hand, **negation** is a ***unary*** operator.

Disjunction

- Let *p* and *q* be propositions.
- The *disjunction* of *p* and *q*, denoted by $p \vee q$, is the proposition "*p* or *q*."
- The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.
- Disjunction is true when at least one of the components is true.
- Disjunction corresponds to English "**OR**".
  - Example: John is brave OR intelligent

Truth Table for Disjunction

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TABLE 3 **The Truth Table for the Disjunction of Two Propositions.**

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Examples of Conjunction & Disjunction

Let,

p : 5 < 9

q : 9 < 7.

Construct the propositions $p \wedge q$ and $p \vee q$.

Solution:

- The conjunction of the propositions p and q is the proposition

  $p \wedge q$: 5 < 9 and 9 < 7

- The disjunction of the propositions p and q is the proposition

  p ∨ q : 5 < 9 or 9 < 7

Question: What are the truth values of $\boldsymbol{p \wedge q}$ *and* $\boldsymbol{p \vee q}$?

Examples of Conjunction & Disjunction (See Example 5 & 6 @ Page 4 & 5)

Consider the following propositions:

p : It is Friday

q : It is raining.

Construct the propositions **p** $\wedge$ **q** and **p** $\vee$ **q.**

Solution:

The conjunction of the propositions p and q is the proposition

p $\wedge$ ***q*** : It is Friday and it is raining.

The disjunction of the propositions p and q is the proposition

p $\vee$ ***q*** : It is Friday or It is raining.

Exclusive Or

- Let p and q be propositions.
- The *exclusive or* of p and q, denoted by $p \oplus q$, is the proposition that is **true** when exactly one of p and q is true and is **false** otherwise.

Truth Table of Exclusive Or

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TABLE 4 **The Truth Table for the Exclusive Or of Two Propositions.**

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

- Let *p* and *q* be propositions.
- The *conditional statement* $p \rightarrow q$ is the proposition "if *p,* then *q*."
- The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.
- In the conditional statement $p \rightarrow q$, ***p*** *is called* ***hypothesis*** *and* ***q*** *is called* ***conclusion***.
- This one is the English usage of "if, then" or "implies".
- The connective $\rightarrow$ is called the 'conditional connective'.
- A **conditional statement** is **also called** an **implication**.

Truth Table for Conditional Statement

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TABLE 5 The Truth Table for the Conditional Statement $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalent Expression of $p \rightarrow q$

- In terms of words, the proposition **p → q also reads**:

  (a) **if p, then q, *or***

  if p, q

  *(The word "**then**" is sometimes omitted in English sentences)*

  (b) **p implies q**

  (c) **p is a sufficient condition for q , *or***

  a sufficient condition for q is p

  (d) **q is a necessary condition for p, *or***

  a necessary condition for p is q

Equivalent Expression of $p \rightarrow q$

(e) **p only if q**

(f) **q if p** , ***or***

q, if p

(g) **q whenever p**

(h) **q when p**

(i) **q unless ¬p**

Remember!

- The **hypothesis** expresses a **sufficient condition**
- The **conclusion** expresses a **necessary condition**
- "***but***" is a logical synonym for "***and***"
- "***when***" / "***whenever***" means the same as "***if***"
- The **hypothesis** is the clause following "**if**"
- The ***conclusion*** is the clause following "***then***"
- "***only if***" clause is the **conclusion**
- ***If*** hypothesis, ***then*** conclusion

Example of Conditional Statement

p = "Cruise ships only go on big rivers."

q = "Cruise ships go on the Hudson."

r = "The Hudson is a big river."

- $p \wedge q$ = "Cruise ships only go on big rivers **and** go on the Hudson."

- $(p \wedge q) \rightarrow r$ = "**If** cruise ships only go on big rivers and go on the Hudson, **then** the Hudson is a big river."

Example 7 (page 7)

- Let p : "Maria learns discrete mathematics" and
  q : "Maria will find a good job."
  Express the statement **p → q** as a statement in English.

Solution:

*"**If** Maria learns discrete mathematics, **then** she will find a good job."*

- There are many other ways to express this conditional statement in English.
  - *"Maria will find a good job **when** she learns discrete mathematics"*
  - *"**For** Maria to get a good job, **it is sufficient for her** to learn discrete mathematics".*
  - *"Maria will find a good job **unless** **she does not** learn discrete mathematics."*

Converse, Contrapositive, and Inverse

- We can form some new conditional statements starting with a conditional statement $p \rightarrow q$

- The **Converse of** $p \rightarrow q$ is the proposition $q \rightarrow p$
- The ***Contrapositive* of** $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$
- The ***Inverse of*** $p \rightarrow q$ is the proposition $\neg p \rightarrow \neg q$
- *Example:*

The given proposition is in the form "q whenever p" such that,
p (hypothesis): It is raining.
q (Conclusion): The home team wins.
Converse: q→p is "If the home team wins then it is raining",
Contrapositive: ¬q→¬p is "If the home team does not win then it is not raining".
Inverse: ¬p→¬q is "If it is not raining then the home team does not win".

Examples of Converse, Contrapositive and Inverse

- ***Converse***: $p \rightarrow q$ ==> $\boldsymbol{q \rightarrow p}$
  - Example: "If it is noon, then I am hungry."

    Converse: "If I am hungry, then it is noon."
- ***Contrapositive***: $p \rightarrow q$ ==> $\boldsymbol{\neg q \rightarrow \neg p}$
  - Example: "If it is noon, then I am hungry."

    Contrapositive: "If I am not hungry, then it is not noon."
- ***Inverse***: $p \rightarrow q$ ==> $\boldsymbol{\neg p \rightarrow \neg q}$
  - Example: "If it is noon, then I am hungry."

    Inverse: "If it is not noon, then I am not hungry."

Bi-Conditional

- Let *p* and *q* be propositions.
- The *biconditional statement* ***p*** $\leftrightarrow$ ***q*** is the proposition **"*p* if and only if *q*."**
- **The biconditional statement *p* $\leftrightarrow$ *q* is true when p and q have the same truth values, and is false otherwise.**
- Biconditional statements are also called "*bi-implications*"

- Question : Which operator is the opposite of $\leftrightarrow$?
- *Answer:* $\leftrightarrow$ has exactly the opposite truth table as $\oplus$.

Truth Table for Biconditional $p \leftrightarrow q$

TABLE 6 The Truth Table for the Biconditional $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example of Biconditional statement

- **Example 10 (Page 9):** Let *p* be the statement " You can take the flight" and *q* be the statement " You buy a ticket".

  What is the statement for $p \leftrightarrow q$?

- **Solution:**

"You can take the flight if and only if you buy a ticket"

How to **Construct** a **Truth Table** for a Compound Proposition?

- At first look at the number of propositions (e.g. *p*, *q*, *r*) in the given compound proposition.
- There will be 2^n number of rows in the truth table, where n is the number of propositions in the compound proposition.
- Draw the table. In the first row, write down the name of propositions (e.g. *p*, *q*, *r*) starting from left/first column.
- Construct the truth table step by step.

Example: Construct a truth table for **(p ⊕ q) ⊕ r**

Solution:

p	q	r	$p \oplus q$	$(p \oplus q) \oplus r$
T	T	T	F	T
T	T	F	F	F
T	F	T	T	F
T	F	F	T	T
F	T	T	T	F
F	T	F	T	T
F	F	T	F	T
F	F	F	F	F

How to determine whether two compound propositions are logically equivalent?

- Two compound propositions are ***logically equivalent*** if they always have the same truth values in the corresponding rows.
- Construct a truth table for the given two compound propositions [in one table]
- **If the truth values of both of the compound propositions are same in the corresponding rows, then they are logically equivalent.**
- If the true values of both of the compound propositions are different in one or more rows, they are not logically equivalent.

Example

- Show that ***p*** $\leftrightarrow$ ***q*** is **logically equivalent** to **(p** $\rightarrow$ **q)** $\wedge$ **(q** $\rightarrow$ **p)**
- **Solution: Since the truth values of both of the compound propositions are same in the corresponding rows, they are logically equivalent**

p	q	p->q	q->p	p<->q	(p->q) ^ (q->p)
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Truth Tables of Compound Propositions

- We can use logical connectives/operators to build up complicated compound propositions involving any number of propositional variables.
- We can use truth tables to determine the truth values of these compound propositions.

- **<u>Example 11 (p. 10):</u> Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$**

Truth Tables of Compound Propositions

TABLE 7 The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are *Logically Equivalent*

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Precedence of Logical Operators

- Negation operator is applied before all other logical operators
- Conjunction operator takes precedence over disjunction operator
- Conditional and biconditional operators have lower precedence
- Parentheses are used whenever necessary

Precedence of Logical Operators

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TABLE 8
Precedence of Logical Operators.

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Logic and Bit Operations

- **bit** ==> ***b***inary dig***it***
- Boolean variable: either **true** or **false**
  - Can be represented by a bit
- **Bit String**: A *bit string* is a sequence of zero or more bits. The *length* of the string is the number of bits in the string.
- Example 20(p.15): 101010011 is a bit string of length *nine*

Bit Operations

- Computer **bit operations correspond to the logical connectives.** By replacing true by a one and false by a zero in the truth tables for the operators $\wedge$, $\vee$, and $\oplus$, the tables shown in Table 9 for the corresponding bit operations are obtained. We will also use the notation *OR, AND, and XOR for the operators* $\vee, \wedge,$ *and* $\oplus$ *respectively, as is done in various programming languages.*

Table for Bit Operations

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TABLE 9 **Table for the Bit Operators *OR*, *AND*, and *XOR*.**

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Bit string and bit operation

DEFINITION 7 A *bit string* is a sequence of zero or more bits. The *length* of this string is the number of bits in the string.

EXAMPLE 12 101010011 is a bit string of length nine. ◀

We can extend bit operations to bit strings. We define the **bitwise *OR***, **bitwise *AND***, and **bitwise *XOR*** of two strings of the same length to be the strings that have as their bits the *OR*, *AND*, and *XOR* of the corresponding bits in the two strings, respectively. We use the symbols $\vee$, $\wedge$, and $\oplus$ to represent the bitwise *OR*, bitwise *AND*, and bitwise *XOR* operations, respectively. We illustrate bitwise operations on bit strings with Example 13.

EXAMPLE 13 Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101. (Here, and throughout this book, bit strings will be split into blocks of four bits to make them easier to read.)

Solution: The bitwise *OR*, bitwise *AND*, and bitwise *XOR* of these strings are obtained by taking the *OR*, *AND*, and *XOR* of the corresponding bits, respectively. This gives us

```
01 1011 0110
11 0001 1101
-------------
11 1011 1111   bitwise OR
01 0001 0100   bitwise AND
10 1010 1011   bitwise XOR
```

◀